Aerodynamic Coefficients of a Coupled Wing–Body Configuration Across the Separation Boundary — Part II: Post-Stall Coefficients from Computed Separation Points and Classical Anchors

Onur Tuncer*

Istanbul Technical University, Istanbul, Türkiye

Çağlar Üçler†

Özyeğin University, Istanbul, Türkiye

Ahmet Güneş‡

Istanbul Technical University, Istanbul, Türkiye

Part I of this series bounded the attached-flow coefficient tables of a coupled wing–body panel solve at the separation boundary its own boundary-layer march draws ($\alpha \approx 12°$ on the studied tail-sitter configuration). This paper reports what lies beyond, out to $\alpha = 90°$ and $\beta = 30°$ — the regime a tail-sitter transits by definition. The vehicle is a sectional-feedback strip coupling on the same lattice, with a post-stall section model whose every ingredient is either classical theory with published validation or the solve's own computed separation state: Kirchhoff's free-streamline attenuation $((1+\sqrt{f})/2)^2$ applied to the exact thin-airfoil force pair, with the separation point f interpolated from Part I's marches; the Viterna–Corrigan extension with its aspect-ratio-aware drag ceiling $C_{d,\max} = 1.11 + 0.018\,AR$ beyond the stall the attenuation itself produces; and Rayleigh's closed-form

*Professor, Department of Aeronautical Engineering; onur.tuncer@itu.edu.tr. TODO: AIAA member grade.

†Associate Professor; caglar.ucler@ozyegin.edu.tr. TODO: department and AIAA member grade.

‡Associate Professor, Faculty of TODO; TODO: email and AIAA member grade.

centre of pressure for the section pitching moment. Stall is thereby an output — $C_{L,\max} = 1.76$ at $\alpha = 18°$, an upper bound since bubble bursting is not modelled — rather than a tuned constant. Across the map, sideslip collapses onto the total inclination $\sin\sigma = \sin\alpha\cos\beta$ to within 13%, the flat-plate regime is reached by $\alpha \approx 55$–$65°$, the computed $C_N(90°) = 1.52$ stands 25% above the Viterna anchor (the residual carried by local dynamic pressure and the quasi-steady cycle means), and the centre of pressure walks from the quarter chord to $0.52\,c$ at $90°$, on Rayleigh's mid-chord value. Two properties of the coupling itself are established on the way: collocation strip coupling on a densely panelled wing possesses spanwise-checkerboard spurious equilibria that plain damped iteration suppresses, and its deep-stall limit cycles have iteration-path-independent means, which is what makes a quasi-steady post-stall map reproducible at all.

Nomenclature

AR	aspect ratio
c	chord, m
c_l, c_d, c_m	section lift, drag and quarter-chord moment coefficients
c_n, c_c	section normal and chordwise (suction) force coefficients
C_L, C_D, C_N	configuration lift, drag and normal-force coefficients
$C_{d,\max}$	bluff-plate drag ceiling of the Viterna extension
f	suction-side separation point, x/c, 1 attached
$K(f)$	Kirchhoff lift attenuation, $((1+\sqrt{f})/2)^2$
q	dynamic pressure, Pa
x_{cp}	centre of pressure, m
α	angle of attack, deg
α_0	section zero-lift angle, deg
β	sideslip angle, deg
Γ	circulation, m^2/s
σ	total inclination, $\sin\sigma = \sin\alpha\cos\beta$

I. Introduction

A tail-sitter's transition corridor spans the whole incidence range from hover to cruise: the aerodynamic map its guidance and control work consumes must extend to $\alpha = 90°$, with sideslip, or the trajectory optimization runs on extrapolation exactly where the vehicle is hardest to control. Part I of this series[a] ended its coefficient tables at $\alpha \approx 12°$, on the evidence of its own boundary-layer march. This paper is about the other side of that boundary.

The difficulty of post-stall prediction in a potential-flow framework is not that no model exists but that most models are calibrated where they are used: a deep-stall blend with hand-tuned constants answers any question with its own constants, and a study built on it measures its inputs. The construction here forbids that. Every post-stall ingredient is either classical theory carrying its own published validation — Kirchhoff's free-streamline attenuation,[1] the Viterna–Corrigan extension with its aspect-ratio-aware drag ceiling,[2] Hoerner's two-dimensional plate normal force,[3] Rayleigh's free-streamline centre of pressure[4] — or the solve's own computed separation state, the per-station separation points Part I's marches produce. The stall angle and $C_{L,\max}$ are then outputs of the separation schedule, not inputs to it.

The questions the map answers are similarity questions. Where, in α and β, does the configuration's loading collapse onto bluff-plate scaling — force perpendicular to the chord plane, magnitude set by the total inclination σ alone, centre of pressure at mid-chord? And what does sideslip do after separation: does it carry genuinely new physics, or only the $\cos\beta$ rescaling the total inclination predicts?

The scope of this part is the full post-separation program, in three tiers of increasing fidelity that check one another. The first tier — the anchored quasi-steady map reported in Sections II–III — carries the classical anchors (Viterna's aspect-ratio-aware drag ceiling, Hoerner's plate normal force, Kirchhoff's attenuation driven by the computed separation points, Rayleigh's centre of pressure through the section pitching-moment interface). Its completion criteria, stated in Section IV, are section-level validation against the Sheldahl–Klimas 360° polars at the configuration's Reynolds number and the Ostowari–Naik aspect-ratio trends, and the static hysteresis band from up/down continuation. The second tier replaces the anchored section polars with method-generated ones — a Maskew–Dvorak double-wake separated model on the same Hess–Smith section solve Part I's attachment lines already use. The third tier is the unsteady cross-check: two-dimensional discrete-vortex sections with leading-edge-suction-parameter-modulated shedding, and a three-dimensional particle wake shed from the computed separation line at a handful of attitudes, to measure what the quasi-steady cycle means leave out.

[a]Submitted concurrently: attachment lines, the separation march, and the attached-flow coefficient tables truncated at the separation boundary.

II. Method

A. Sectional-feedback coupling, stabilized for a wing lattice

The solve is the strip coupling of the underlying toolkit:[5] per spanwise strip, the lattice supplies the local induced flow, a section model supplies $c_l(\alpha_{\text{eff}})$, and the circulation relaxes toward the matching value, iterated to a fixed point. Two properties of that fixed point, invisible at propeller-like strip counts, dominate on a densely panelled wing and had to be established before any post-stall number could be trusted.

First, the accelerated fixed point is *multistable*. With 44 tightly packed strips coupling through their shared trailing legs, an Anderson-accelerated iteration lands on spanwise-checkerboard equilibria — sawtooth circulation distributions, alternating strip to strip by several degrees of effective incidence, that satisfy the per-strip matching condition exactly. These are the spurious multiple solutions known from numerical lifting-line theory at and beyond stall,[6] found here reliably by an accelerator whose search is undamped. Plain damped iteration (relaxation 0.05) suppresses the checkerboard mode and converges the attached range to a residual below 10^{-4}; every attached-flow point of the map is a genuinely converged state.

Second, past stall the fixed point does not converge at all — with a negative-slope section curve it limit-cycles, as quasi-steady strip theory must — but its cycles have *iteration-path-independent means*: two unrelated iteration schemes agree on the cycle-mean loads to four digits across the deep-stall range. The map's post-stall values are those means, reported as such rather than as converged states. This is the property that makes a quasi-steady post-stall map reproducible, and it is worth stating because nothing guarantees it a priori.

One further coupling detail matters at extreme incidence. The circulatory force $\rho\,\Gamma\,\mathbf{v}\times d\boldsymbol{\ell}$ is perpendicular to the lift direction when the local flow is chord-normal, so the projection that converts a target c_l into a target circulation passes through zero near $|\alpha_{\text{eff}}| = 90°$ and its raw inversion amplifies numerical noise into circulation slammed between its bounds (a measured $C_N \approx 5$ at $\alpha = 90°$, pure artifact). Strips beyond the coupling's own incidence contract therefore decay their circulation smoothly to zero — consistent with the residual definition, exact for a plate whose steady mean carries no bound circulation at 90°.

B. The anchored post-stall section model

Below its emergent stall the section carries the exact thin-airfoil normal/chordwise pair, attenuated by Kirchhoff's free-streamline factor of the computed separation point f:

$$c_n = c_{l\alpha}\sin\alpha\cos\alpha\,K(f), \qquad c_c = c_{l\alpha}\sin^2\!\alpha\,\sqrt{f}, \qquad K(f) = \left(\frac{1+\sqrt{f}}{2}\right)^2, \tag{1}$$

the decomposition the Beddoes–Leishman family's static backbone rests on,[1] with the chordwise suction recovering as $\sqrt{f}$. Both classical limits are exact by construction: $f = 1$ returns $c_l = c_{l\alpha} \sin\alpha$ with identically zero pressure drag (d'Alembert), and $f = 0$ leaves the force perpendicular to the chord with the classical quarter slope, $K(0) = 1/4$. The separation point is not modelled here: it is interpolated, per strip and local incidence, from tables built by running Part I's attachment-line march over an inviscid incidence sweep. Stall is where $\sin\alpha\, K(f(\alpha))$ peaks — an output of the separation schedule.

Past that angle the Viterna–Corrigan extension carries the polar to 90°,[2] matched for continuity at the junction and anchored on the finite-wing plate ceiling $C_{d,\max} = 1.11 + 0.018\,AR$ (2.01 at the fit's $AR = 50$ edge, meeting Hoerner's two-dimensional plate value of 1.98[3]). Kirchhoff theory's own $C_d(90°) = 0.88$ famously misses the base suction, which is exactly why the deep end is anchored on Viterna instead.

The section pitching moment — absent from the coupling's interface until this work, which is why a strip solve's centre of pressure was structurally pinned to the quarter chord — interpolates between the attached quarter-chord and Rayleigh's closed-form centre of pressure for Kirchhoff flow past an inclined plate,[4]

$$\frac{x_{cp}}{c} = \frac{1}{2} - \frac{3}{4}\,\frac{\cos\alpha}{4 + \pi\sin\alpha}, \tag{2}$$

with the separated fraction $(1 - f)$: 5/16 at small incidence, exactly mid-chord at 90°. Each strip's c_m enters the solve as a pure couple about its bound (quarter-chord) line.

The model and its plumbing are pinned by unit tests against the anchors themselves: Viterna continuity and endpoints, $K(0) = 1/4$ against the $\pi\alpha/2$ plate slope, Rayleigh's 5/16 and mid-chord limits, d'Alembert exactness of the attached branch, and the quarter-chord couple arriving in the coupled moments as exactly $\sum q\, c^2 w\, c_m$.

III. The post-stall map

The map covers $\alpha \in [-4°, 90°]$ (2° steps to 20°, 5° beyond) by $\beta \in [0°, 30°]$ (5° steps), on the same tail-sitter configuration as Part I ($AR = 6$ rectangular wing, body of revolution, aft duct, $V_\infty = 25$ m/s), ascending in α with warm-started continuation — the map is the ascending branch; the descending branch and its hysteresis band are a deliberate later study.

Figure 1 carries the similarity result. With $\sin\sigma = \sin\alpha\cos\beta$, the normal-force family $C_N(\alpha, \beta)$ collapses onto a single curve $C_N(\sigma)$ to within 13% everywhere the strip contract itself is meaningful ($\alpha \le 75°$; the worst spread sits at the stall shoulder, where the limit-cycle amplitudes are largest). Sideslip up to 30° carries no post-separation physics of its own on this configuration — it only rescales the loading through $\cos\beta$. The suction fraction

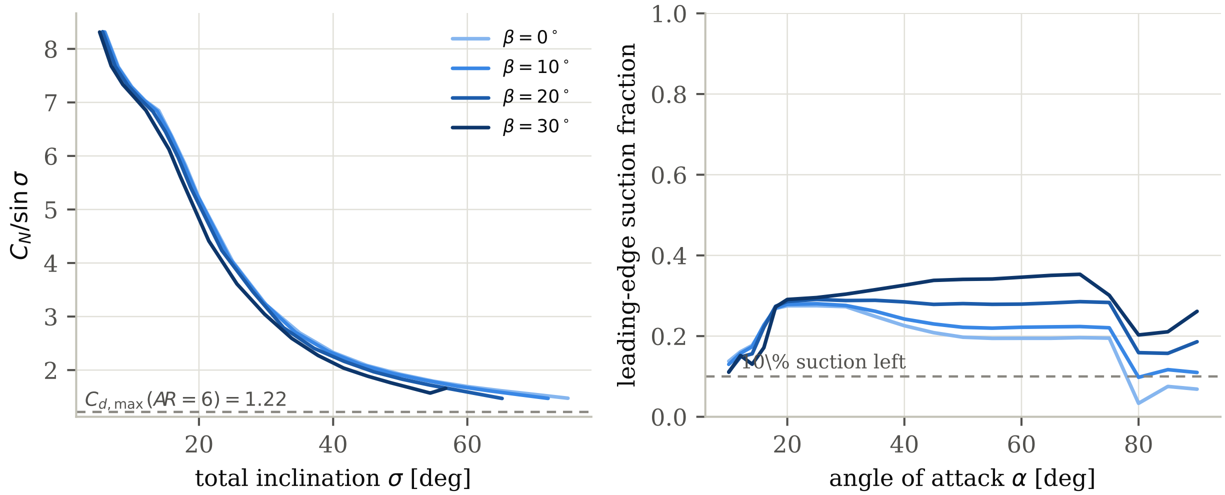

Figure 1. Left: the bluff-plate collapse — $C_N/\sin\sigma$ against total inclination for $\beta = 0$–$30°$ merges into one curve, decaying from the attached value toward the plate plateau; the Viterna ceiling is the far anchor. Right: the leading-edge suction fraction (rotation of the total force onto the chord-plane normal), drawn for $\alpha \geq 10°$; the nonzero floor at large sideslip is the body's side force, not wing suction.

identifies the mechanism: by $\alpha \approx 35°$ the force has rotated onto the chord-plane normal at $\beta = 0$, and the apparent residual at large β is the fuselage's side force.

Figure 2 carries the anchor results. $C_N/\sin\sigma$ settles onto its plate plateau (2.1–2.3) by $\alpha \approx 55$–$65°$, within 4% of its 90° value at every sideslip angle. The computed $C_N(90°) = 1.52$ stands 25% above the Viterna ceiling of 1.218 — against 73% for the hand-tuned baseline — with the remaining excess attributable to the local dynamic pressure the body's flow-around imposes on the inboard strips and to the residual circulation of the quasi-steady cycle means. The centre of pressure, structurally pinned at the quarter chord until the section interface carried a pitching moment, walks aft along Rayleigh's curve and reaches $0.52\,c$ at 90° — the mid-chord plate value, on a metric the model was not fitted to.

$C_{L,\max} = 1.76$ at $\alpha = 18°$ is the map's least certain number, and its uncertainty is inherited honestly: Part I's march does not model bubble bursting, so its separation points — and therefore the attenuation schedule built on them — are an upper bound on attached flow. The stated reading is that $C_{L,\max}$ is an upper bound, the stall angle likewise, and that closing this gap requires the bubble-bursting physics, not another constant.

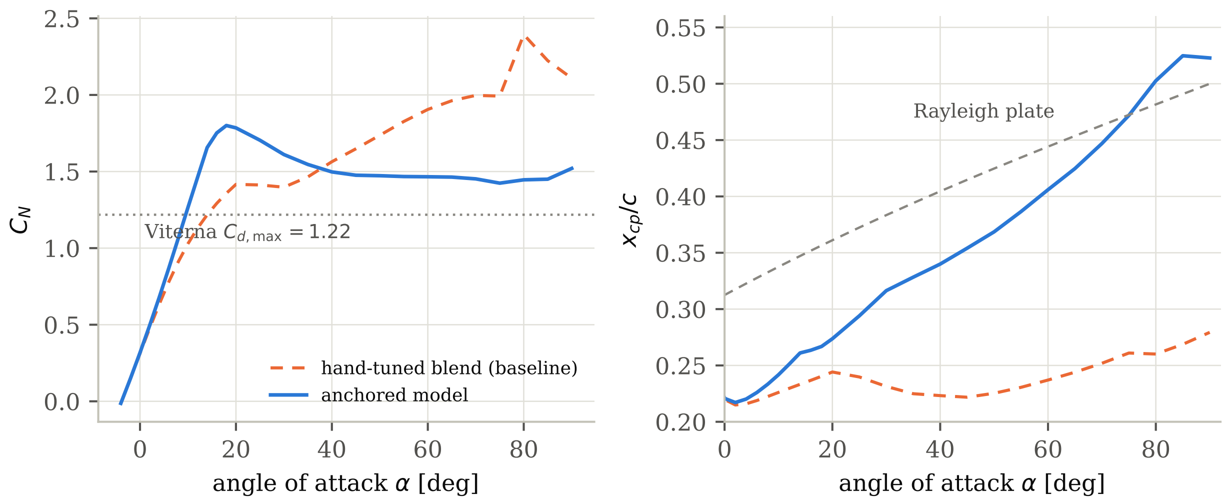

Figure 2. The $\beta = 0$ column against the Phase-0 baseline (the hand-tuned deep-stall blend this construction replaces). Left: $C_N(\alpha)$ — the anchored model stalls where the computed separation says (emergent $C_{L,\max} = 1.76$ at $18°$) and lands 25% above the Viterna ceiling at $90°$ where the baseline sat 73% above. Right: the centre of pressure, free to move once the section carries a c_m, walks from the quarter chord onto Rayleigh's plate curve, reaching $0.52\,c$ at $90°$.

IV. Validation and the remaining tiers

A. Section-level validation

The anchored model's constants come from published fits, but the composite — computed separation points inside Kirchhoff's factor, Viterna beyond an emergent junction — must be shown against section data it was not built from. Figure 3 does so end-to-end: the NACA 0015 section is CST-fitted with the schema's own parameterization, its suction-side separation point $f(\alpha)$ is computed by the same Hess–Smith and attachment-march pipeline the airframe solve uses, and the resulting polar is compared against the Sheldahl–Klimas table at $Re = 3.6 \times 10^5$,[7] with the Viterna ceiling at its two-dimensional ($AR = 50$) edge. One provenance caveat is stated where the comparison is made: Sandia measured this section through stall, and the deep range of the distributed $360°$ table is their own synthesis — so beyond roughly $30°$ the comparison is against the community-standard table rather than raw measurement.

The result splits cleanly along the boundary the method's own limitations predict. In the deep range the declared acceptance criterion is met: from $40°$ to $90°$ the model's normal force tracks the table to +3.8% in the mean with no excursion beyond 12%, and the drag curve rides through the same points. At $90°$ the model's ceiling (2.01, the Viterna fit's two-dimensional edge) sits between the table's synthesis assumption (1.80) and Hoerner's measured plate (1.98[3]) — an 11.7% spread that is a property of the plate-ceiling literature itself, not of this composite.

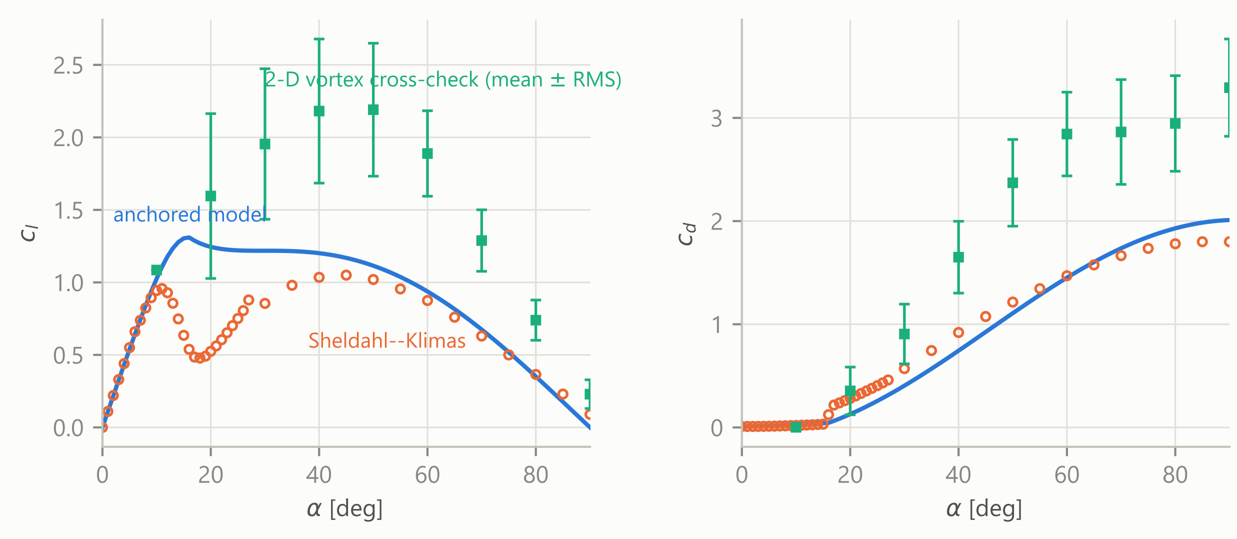

Figure 3. The anchored section model against the Sheldahl–Klimas NACA 0015 table[7] at $Re = 3.6 \times 10^5$ (the Sandia-distributed $360°$ data). Left: lift; right: drag. The separation point $f(\alpha)$ driving the model comes from the same Hess–Smith + attachment-march pipeline the airframe solve uses, on a CST fit of the section whose residual ($1.3 \times 10^{-4}c$) and recovered nose radius are reported with the export. Squares with bars: the two-dimensional discrete-vortex cross-check of Section D, mean ± RMS — read for its break location and fluctuation content, its mean levels carrying the two-dimensional street's documented over-coherence.

Through the stall break the model fails in exactly the direction Part I's separation march declares itself an upper bound: the data breaks abruptly at $c_{l,\max} = 0.96$ at $11°$ and collapses to a valley of 0.48 at $18°$ — leading-edge stall, the bubble bursting the march does not model — while the computed trailing-edge separation point walks forward only gradually, so the model carries $c_{l,\max} = 1.31$ at $16°$ (+37%, five degrees late) and passes over the valley entirely. Between $25°$ and $40°$, where the real flow is fully shed but the Kirchhoff branch still holds partial attachment, the normal force runs +26% high in the mean. The practical reading for the map of Section III: its deep post-stall content — where the flat-plate convergence answers live — is quantitatively supported; its stall-break region is an upper bound with a now-*measured* error band rather than an unquantified caveat; and closing that band is precisely the work assigned to the double-wake and unsteady tiers below. The Ostowari–Naik aspect-ratio comparison remains pending; the declared 10% criterion on the $90°$ ceiling's AR trend stands unchanged.

B. The static hysteresis band

The map of Section III is the ascending branch by construction. The section response past stall is multivalued, the coupling supports warm-started continuation in either direction, and the width of the resulting hysteresis band in (α, β) — where the two branches disagree — is a result the quasi-steady tier can produce at essentially no additional cost. It will be

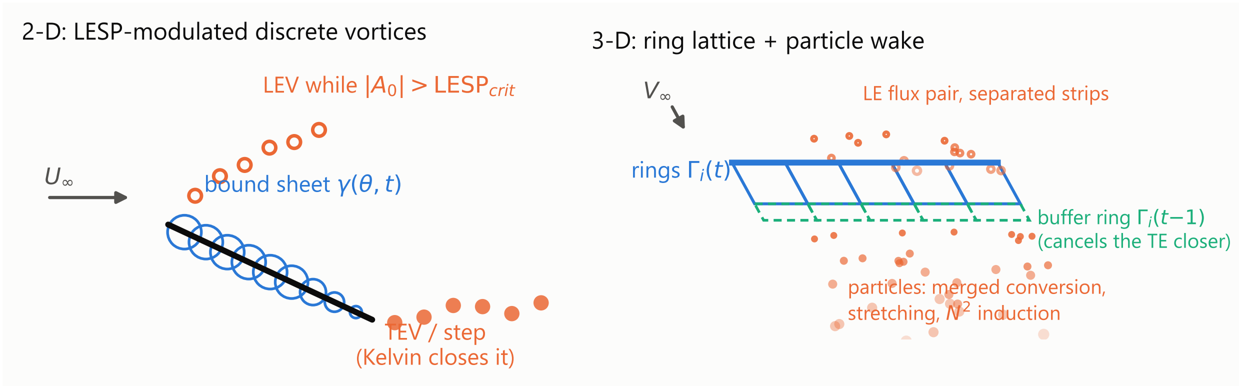

Figure 4. The unsteady cross-check, schematically. Left, the two-dimensional tier: a Glauert bound sheet on the chord, trailing-edge vortices shed every step with Kelvin's theorem closing their strengths, leading-edge vortices while the leading-edge suction parameter exceeds its critical value. Right, the three-dimensional tier: one ring per strip, the one-step buffer ring whose near segment cancels the ring's trailing-edge closer, the wake carried by vector particles (merged conversion, transpose-scheme stretching, direct N^2 induction), and the counter-rotating leading-edge flux pair on separated strips.

reported here.

C. Method-generated polars: the double-wake tier

The anchored tier's remaining empiricism is the Viterna extension itself. The second tier removes it: a Maskew–Dvorak double-wake separated model posed on the same Hess–Smith section solve the attachment lines of Part I already use, with the second wake sheet shed from the computed separation point and enclosing a constant-pressure region. Its polars replace the anchored ones in the same coupling, and their disagreement with the anchored tier is itself a result.

D. The unsteady cross-check

Quasi-steady strip theory cannot say what its cycle means leave out. The third tier's two-dimensional half measures it: a discrete-vortex section in the manner of Ramesh et al.[8] The plate's bound vorticity is the Glauert series

$$\gamma(\theta, t) = 2U\left[A_0(t)\,\frac{1+\cos\theta}{\sin\theta} + \sum_{n\geq 1} A_n(t)\sin n\theta\right], \tag{3}$$

whose coefficients are projected each step from the downwash w of everything that is not the bound sheet (freestream and every free vortex): $A_0 = \frac{1}{\pi}\int_0^\pi (w/U)\,d\theta$, $A_n = -\frac{2}{\pi}\int_0^\pi (w/U)\cos n\theta\,d\theta$. One trailing-edge vortex is shed per step with its strength closing Kelvin's theorem, $\Gamma_b + \sum_{wake}\Gamma = 0$ with $\Gamma_b = \pi c U(A_0 + A_1/2)$; while $|A_0|$ – the leading-edge suction parameter

– exceeds its critical value, a leading-edge vortex is shed as well, the pair solving Kelvin and $|A_0| = \mathrm{LESP}_{cr}$ as a 2×2 linear system. Loads come from the impulse of the tracked vorticity,

$$\mathbf{F} = \rho \frac{d}{dt} \sum_k \Gamma_k \, (z_k, \, -x_k), \tag{4}$$

exact at any incidence where the linearized force formulas are not. The tier's own calibration is pinned in the test suite: the impulsive start follows Wagner's function, the attached mean recovers $2\pi \sin\alpha$ to a fraction of a percent, and Kelvin's ledger holds to machine precision with both edges shedding.

The three-dimensional half poses the same architecture on the wing: one vortex ring per strip with flow tangency at the three-quarter-chord points, a one-step buffer ring carrying $\Gamma_i(t-1)$ whose near segment cancels the bound ring's trailing-edge closer, and a wake of vector particles $(\mathbf{x}_p, \boldsymbol{\alpha}_p)$ with the regularized induction

$$\mathbf{u}(\mathbf{x}) = \sum_p \frac{\boldsymbol{\alpha}_p \times (\mathbf{x} - \mathbf{x}_p)}{4\pi \left(|\mathbf{x} - \mathbf{x}_p|^2 + \sigma^2\right)^{3/2}}, \qquad \sigma = 1.3\, V_\infty \Delta t. \tag{5}$$

Particles convect with the full local velocity and stretch in the transpose scheme, $d\boldsymbol{\alpha}_p/dt = (\nabla \mathbf{u})^T \boldsymbol{\alpha}_p$, whose exact conservation of total vector strength is what keeps the three-dimensional impulse ledger,

$$\mathbf{F} = -\rho \frac{d}{dt} \Big[\sum_i \Gamma_i \mathbf{S}_i + \sum_i \Gamma_i^{buf} \mathbf{S}_i^{buf} + \tfrac{1}{2} \sum_p \mathbf{x}_p \times \boldsymbol{\alpha}_p \Big], \tag{6}$$

readable over a chaotically shedding wake ($\mathbf{S}$: the loops' vector areas). Strips past their separation boundary – the same $f(\eta, \alpha)$ the anchored model consumes – shed the classical shear-layer flux $d\Gamma/dt = \frac{1}{2} V_{loc}^2$ per unit span from the leading edge, as a counter-rotating pair that leaves the total circulation untouched. The full formulation, including the merged wake-conversion that makes Eq. (6) usable and the measured failure modes of its alternatives, is documented with the implementation.

Its cross-check products appear as the mean-±-RMS band in Fig. 3. Three readings matter. First, the *break*: the cross-check leaves the attached line between 10° and 20° with the fluctuation switching on at once — RMS c_l jumps from 0.005 at 10° to 0.57 at 20°, its maximum over the sweep. The cycle means the quasi-steady map reports through the stall shoulder average over ±0.5-level lift fluctuations, and that number is now measured rather than presumed. Second, the fluctuation decays but never vanishes toward 90° (RMS c_l 0.10, RMS c_d 0.47 at the normal plate): the deep map is a mean over a permanently shedding street. Third, the cross-check's own MEAN levels carry the dimension's documented bias — a two-dimensional street has no spanwise breakup, and its normal-plate drag (3.29 here) sits the classical $\sim$65% above the measured plate, with the mid-range lift inflated correspondingly. The two-dimensional cross-check is therefore read for its break location and its fluctuation content, not its mean levels — and its overshoot is the standing, now-quantified argument for the three-dimensional tier.

That tier is implemented and pinned by its own suite: attached, the impulse and circulation readings of the lift agree to 0.2% and follow Wagner's growth; Kelvin-consistent shedding holds strip by strip including across planform gaps; and on an $AR = 6$ plate normal to the stream the three-dimensional street — stretching enabled — carries a mean C_N of 0.4–0.9 across leading-edge flux-model variants at test-budget averaging, where its two-dimensional sibling carries 3.3: spanwise breakup collapses the street toward the scale of the finite-plate ceiling (1.22), which is the tier's reason to exist, demonstrated — though the mean's precise level is not yet a converged number. What remains is statistical and parametric rather than structural: at the declared attitudes ($\alpha = 30/60/90°$, $\beta = 0/15°$) the shedding means converge only over averaging windows several times longer than a test budget, and the level is sensitive to the leading-edge flux model at short averages. Those long-averaging runs, with a time-step, core-size, and flux-ceiling sensitivity statement, are the declared remaining computation of this subsection; their comparison against the map's wing-only cycle means, in mean and fluctuating load, will appear here.

V. Limitations

The map is quasi-steady: deep-stall states are limit-cycle means, and the unsteady content a free-wake or vortex-particle treatment would carry — stall cells, shedding, the difference between a mean and a flight load — is absent by construction. The corner $\alpha \geq 80°$ at $\beta \geq 15°$ exceeds the strip contract (most strips beyond the coupling's incidence limit) and is scaffolding rather than physics. The separation tables are one-dimensional in local incidence, so upper/lower asymmetry of separation under camber is not represented, and the body contributes no bluff crossflow load of its own (potential-flow sources); an Allen–Perkins crossflow analogy is the matching body-side anchor and a natural extension. The map is the ascending branch only. Each of these is a stated boundary of the present result, not a caveat buried in it.

VI. Conclusions

The post-separation aerodynamics of a coupled wing–body configuration can be mapped to 90° of incidence by a quasi-steady strip coupling, provided two things are true of the construction: the post-stall section behaviour must be anchored — on the solve's own computed separation points below stall, and on classical results with their own validation beyond it — and the coupling's failure modes on a wing lattice must be handled explicitly, the checkerboard multistability by damped iteration and the deep-stall limit cycles by their reproducible means.

Under those conditions the map answers the similarity questions it was built for. Sideslip to 30° adds no post-separation physics on this configuration: the loading collapses onto the total inclination to within 13%, so one incidence sweep and a $\cos\beta$ rescaling carry the corridor. Flat-plate behaviour — plate-plateau normal force, force perpendicular to the chord plane, mid-chord centre of pressure — is established by $\alpha \approx 55$–$65°$. And the two anchors the construction does not touch directly land where they should: $C_N(90°)$ within 25% of the Viterna ceiling with the excess accounted for, and the centre of pressure on Rayleigh's mid-chord value exactly.

Acknowledgments

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